Rule-Based AI Chatbot Using Python

Project README • AI & ML Virtual Internship • Task 5

> **Project Status:** Completed
>
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Project Description

This project is a simple Rule-Based AI Chatbot developed using Python. It interacts with users and provides predefined responses based on their input text.

Objective

To understand basic AI chatbot functionality using Python conditional statements without requiring Machine Learning libraries.

Technologies Used

- Python 3
- VS Code

Features

- Responds to user greetings.
- Answers simple predefined questions.
- Recognizes user messages dynamically.
- Provides instant predefined responses.
- Safely exits when the user types "bye".

Python Concepts Used

- Input and Output handling (`input()` & `print()`)
- If-Elif-Else conditional statements
- While Loop execution flow
- Break Statement for clean exits
- String Functions (`lower()` and `strip()`)

How to Run the Project

1. Install Python 3 on your system.
2. Open your project folder in VS Code.
3. Create a file named `chatbot.py`.
4. Copy the Python chatbot source code into the file.
5. Open the integrated terminal.

6. Run the following command:

```
python chatbot.py
```

Sample Interaction

```
You: hello
Bot: Hello! Welcome.

You: how are you
Bot: I'm fine. Thank you!

You: what is your name
Bot: I am a Simple AI Chatbot.

You: what can you do
Bot: I can answer simple questions.

You: thank you
Bot: You're welcome!

You: bye
Bot: Goodbye! Have a nice day!
```

Conclusion

This project demonstrates a basic Rule-Based AI Chatbot implemented in Python. It helps beginners understand how conversational flow and rule-based logic work using fundamental conditional statements and loops.

Status: Completed Successfully